王思妲

Sarah

第二章　函数的导数(Derivative)

2.1　导数的定义(The Definition of Derivative)

Multiple Choice Questions

$\frac{\Delta y}{\Delta x} = \frac{f(-1)-f(2)}{-1-2} = \frac{1-2}{-3} = \frac{1}{3}$

C　1. Which of the following values is the average rate of change of $f(x)=\sqrt{x+2}$ over the interval $[-1,2]$?

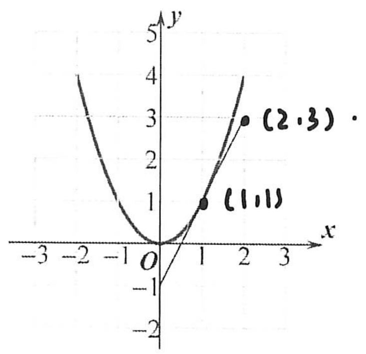
(A)-4　　(B)$-\frac{1}{3}$　　(C)$\frac{1}{3}$　　(D)3

C　2. Let $f(x)=4-3x$. Which of the following is equal to $f'(0)$?　$f'(0)=\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$

(A)4　　(B)-4　　(C)-3　　(D)4

$=-3$

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1)=($ D $)$.



(A)0　　(B)$\frac{1}{2}$　　(C)1　　(D)2

A　4. Let $f(x)=kx^3-2x,(k\neq 0)$. The average rate of $f(x)$ over the interval$[-2,-1]$ is 3, find k.

(A)$\frac{5}{7}$　　(B)$\frac{1}{7}$　　(C)$-\frac{1}{7}$　　(D)$-\frac{5}{7}$

C　5. **Caculator** Let $y=\tan e^{x}-x^2+1$. Find $\frac{dy}{dx}\big|_{x=0.5}=$

(A)-12.056　　(B)0　　(C)271.066　　(D)273.066

C　6. **Caculator** The graph of $y=\sqrt{x^2+1}-3x$ crosses the x-axis at one point in the interval $[0,1]$. What is the slope of the graph at this point?

(A)0.354　　(B)-0.001　　(C)-2.666　　(D)0

B 7. The population of insects at time t days is modeled by the function P with first derivative $P'(t)=0.1t^2+10t+120$. Which of the following statements is true?

~~(A)~~At time $t=10$, the population of insects is 230. 320

(B) At time $t=10$, the population of insects is increasing at a rate of 230 insects per day.

~~(C)~~At time $t=10$, the population of insects is ~~decreasing~~ at a rate of 230 insects per day.

~~(D)~~At time $t=10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

A 8. If the line tangent to the graph of a function g at the point (1,4) passes through the point(−2, −5), what is the slope of this line?

(A)3✓ (B)−3 (C)$\frac{1}{3}$ (D)$-\frac{1}{3}$

Free Response Questions

9. Let $f(x)=x^2-2x$

(a)Find the average rate of change of f over [1,3].

(b)Find $\frac{f(2+h)-f(2)}{h}$.

(c)Find the instantaneous rate of change of f at $x=2$.

(a) $\frac{\Delta y}{\Delta x}=\frac{f(1)-f(3)}{1-3}=\frac{-4}{-2}=2$

(b) $\frac{f(2+h)-f(2)}{h}=\frac{(2+h)^2-2(2+h)}{h}$

$=h+2$

(c) $f'(2)=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$

$=2$

10. Use the definition of derivative to find the derivative of $f(x)=x^3$, then find $f'(2)$.

$$f'(x)=\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

$$=\lim_{\Delta x\to 0}\frac{(x+\Delta x)^3-x^3}{\Delta x}$$

$$=\lim_{\Delta x\to 0}\frac{\cancel{x^3}+3x^2\Delta x+3x(\Delta x)^2+\Delta x^3\cancel{-x^3}}{\Delta x}$$

$$=\lim_{\Delta x\to 0}3x^2+3x\Delta x+(\Delta x)^2$$

$$=3x^2$$

$$f'(2)=3\times4=12$$